

A7: Mass Production (MP)

Report

Team Bob the Builders: Will Castro, Jos Moresi, Li Arditi, and Sam Eppley



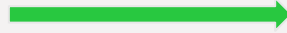
Everything has changed

Low carbon steel



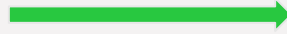
Stainless steel

2-part tube and sheet stock
vessel TIG weld together



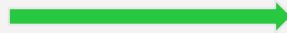
1-part sheet stock vessel deep
draw stamped

Tube and sheet stock plunger
TIG welded together



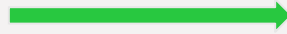
Spin formed plunger with tabs,
laser welded ring

Hand-cut filter



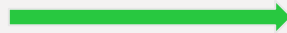
Punched filter

Wire EDM and waterjet cut
tube and sheet stock



Sheet stock punched with die

Lathe and TIG welding fixtures
and tooling

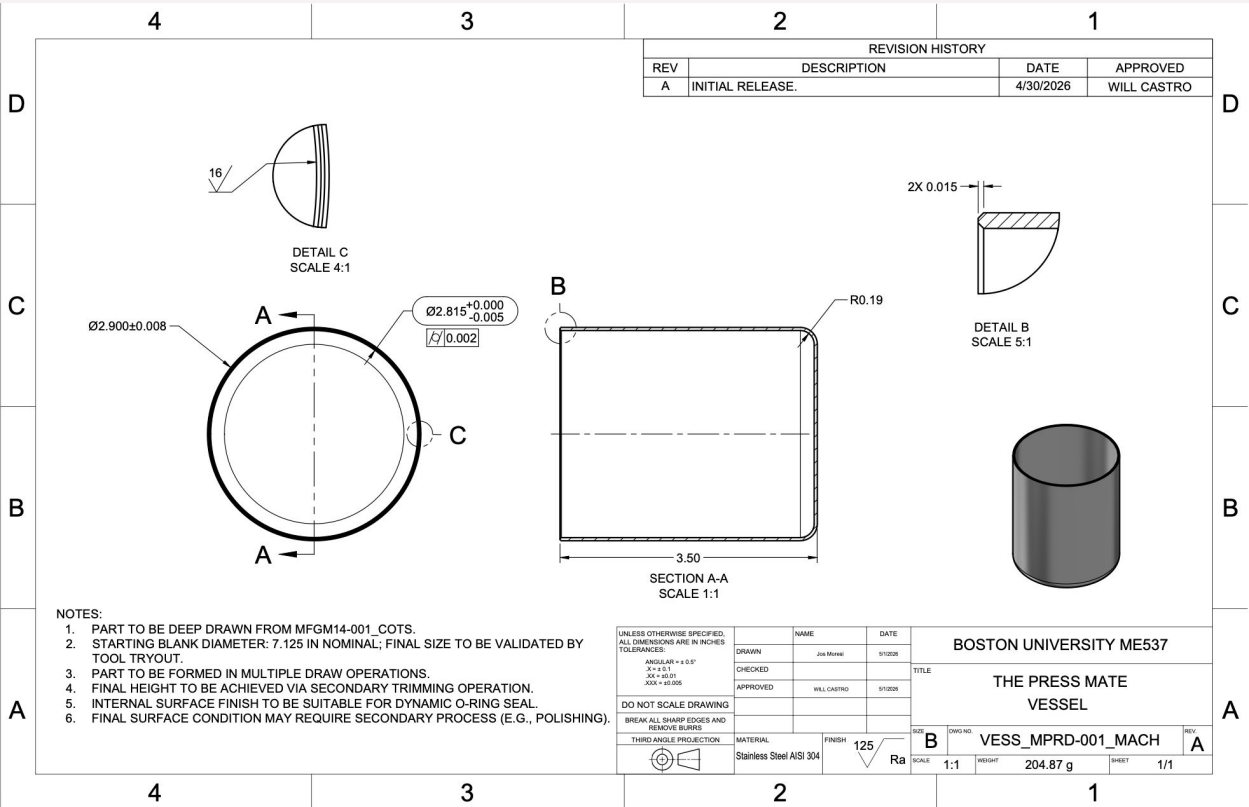


Punches, dies, laser welding, spin
forming tooling

Vessel

The vessel will be deep draw stamped out of a sheet of stainless steel

- Now 1 part instead of 2 - no welding
- No need to turn inner or outer diameters

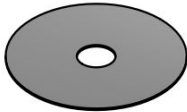


Plunger

Plunger body from cut stainless steel sheets and spin formed

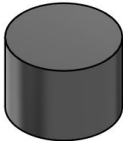
- Tabs are part of the sheet and in the spin forming process
- Plunger is now just 3 part
- Ring and mesh is still welded on using a linkage roller to consistently turn plunger while welding
- Now just 3 parts instead of 5

Filter is stamped out of a sheet of mesh with a die



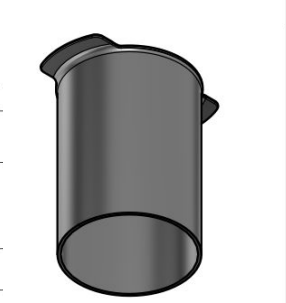
BOSTON UNIVERSITY ME537					
TITLE THE PRESS MATE PLUNGER BLANK					
SIZE B	DWG NO. MFGM15-001_COTS				REV. A
SCALE 1:2	WEIGHT 361.01 g	SHEET 1/1			





BOSTON UNIVERSITY ME537					
TITLE THE PRESS MATE FILTER PUNCH					
SIZE B	DWG NO. TOOL10-001_MACH				REV. A
SCALE 1:1	WEIGHT 1525.89 g	SHEET 1/1			

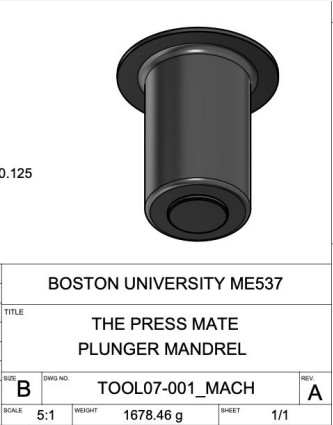
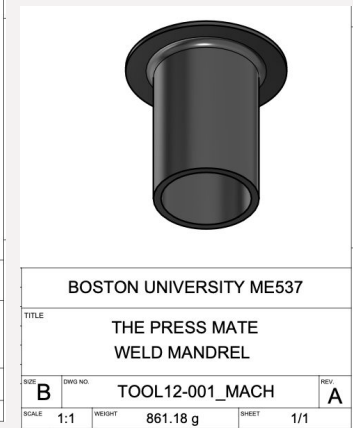
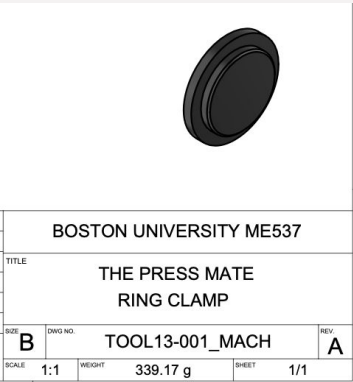
BOSTON UNIVERSITY ME537					
TITLE THE PRESS MATE PLUNGER SPIN					
SIZE B	DWG NO. PLNG_FORM-001_MACH				REV. A
SCALE 1:1	WEIGHT 343.88 g	SHEET 1/1			



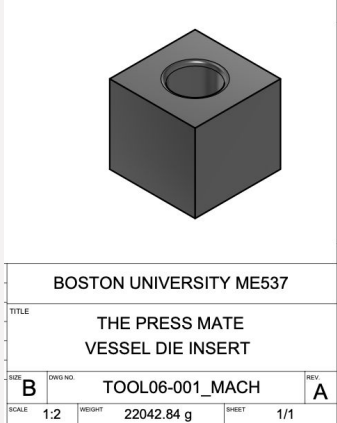
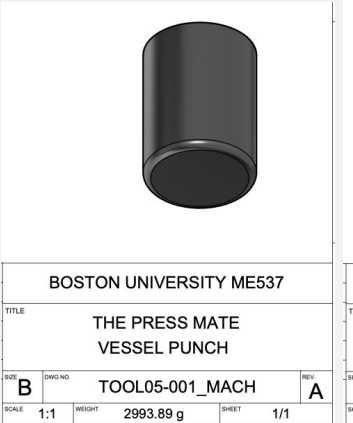
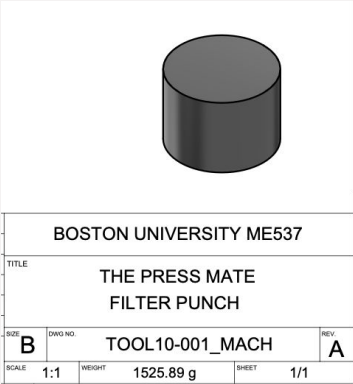
BOSTON UNIVERSITY ME537					
TITLE THE PRESS MATE PLUNGER FINAL					
SIZE B	DWG NO. PLNG_FINL-001_MACH				REV. A
SCALE 6:1	WEIGHT 281.30 g	SHEET 1/1			

Tooling

Laser
Welding
tools



Dies/
Punches



Spin forming
tools

